Zahid Faqih Alim Rabbani

zafar2154.ez@gmail.com | +62 822-5825-8124 | Jakarta, Indonesia
LinkedIn | GitHub | Portfolio

SUMMARY

Electrical Engineering graduate (GPA 3.74) from UPN Veteran Jakarta with hands-on experience in embedded systems, IoT, control systems, machine learning, and mobile/web development. Built a portable IoT water quality monitoring system with Fuzzy Mamdani logic (95.83% accuracy, published in the UPNVJ repository) and led a software engineering division at KSM IoT, receiving the Best Division Head award. Has taught 150+ students in the Mechatronics Laboratory and enjoys turning ideas into practical, working systems.

EDUCATION

Universitas Pembangunan Nasional "Veteran" Jakarta **2022 – 2026**
Bachelor Degree in Electrical Engineering | GPA 3.74 / 4.00

WORK EXPERIENCE

Mechatronics Laboratory Assistant **Mar 2025 – Jul 2026**
Electrical Engineering, UPN Veteran Jakarta

- Taught PLC fundamentals and ladder diagram programming to **150+ electrical engineering students** across **10 practicum sessions per semester**.
- Designed and wrote **Module 1** of the practicum curriculum every semester.

Intern, Protection Division **Jul 2025 – Aug 2025**
PLN

- Studied electrical protection systems, occupational health and safety (K3), and the working environment at PLN.

Algorithm and Programming Laboratory Assistant **Mar 2024 – May 2024**
Electrical Engineering, UPN Veteran Jakarta

- Designed the learning curriculum and guided **80+ electrical engineering students** in C and Python fundamentals, including Matplotlib and Pandas for data analysis.

PROJECTS

Educational Website for Senior High School **Sept 2026**

- Developed a web platform with **60+ books**, exam questions with answers, and interactive simulations for high school students.

Portable IoT-Based Drinking Water Quality Monitoring System | *Undergraduate Thesis* **Jan 2026 – Aug 2026**

- Built a portable device on **ESP32** measuring **turbidity, pH, TDS, air temperature, and water temperature**, with GPS-based spatial mapping of measurements.
- Implemented **Fuzzy Mamdani** logic to assess water quality, achieving **95.83% accuracy**.
- Developed the Android application with **Kotlin and Jetpack Compose**, and designed and 3D-printed the device enclosure using **Fusion 360**.

Anxiety Level Detection from Heart Rate using mmWave Radar **Jul 2026**

- Developed a contactless device that detects a person's anxiety level from heart rate using an mmWave radar sensor.

Autonomous Surface Vessel | *KKI 2024 Finalist* **Sep 2024 – Oct 2024**

- Built an autonomous vessel with **NVIDIA Jetson Nano** and Arduino controllers; implemented **YOLO and OpenCV** for computer vision.
- Integrated a camera, Pixhawk, and HC-SR04 ultrasonic sensor for navigation and obstacle avoidance; built a PHP web-based monitoring system.

Simple Automatic Vacuum Cleaner **Apr 2024**

- Led the project from planning to execution: Arduino with fuzzy-logic control, DHT11 and HC-SR04 sensors, programming and hardware assembly.

Rock Paper Scissors Image Classification **Mar 2024**

- Trained a CNN with data augmentation and transfer learning (TensorFlow, Keras, Google Colab), reaching **98% test accuracy**.

Bathtub Temperature Control **Dec 2023**

- Built an IoT temperature monitor with ESP32, DHT22, ThingSpeak, and a mobile app made in MIT App Inventor.

ORGANIZATIONAL EXPERIENCE

KSM Internet of Things, UPN Veteran Jakarta **Dec 2023 – Jan 2026**

Head of Software Engineering Division (Jan 2025 – Jan 2026) | Staff Software Engineer (Dec 2023 – Dec 2024)

- Led a team of **5 staff**; authored a **10-module syllabus** to train them in software and IoT development. Received the **Best Division Head award**.
- Delivered the KSM IoT UPNVJ website, the web dashboard for the autonomous surface vessel, and a website with machine learning for an automated urine system.

Heyjong Community **Jan 2026 – Jul 2026**

Head of Event Organizer Division

- Led **11 staff** in running weekly community events and **4 major events**, including community service and science teaching for children.

AWARDS & CERTIFICATION

- **Best Head Division Award**, KSM Internet of Things UPNVJ
- **Finalist, KKI 2024** (Autonomous Surface Vessel)
- **DBS Foundation Coding Camp 2024** (Dec 2024): Applied Machine Learning, Machine Learning Intermediate and Basic, Python, Data Visualization
- **Code Generation and Optimization Using IBM Granite**, IBM SkillsBuild (Oct 2025)

SKILLS

Embedded & IoT: ESP32, Arduino, Raspberry Pi, NVIDIA Jetson Nano, Pixhawk, sensor integration, MQTT, Rest API, Node Red
Control & Intelligent Systems: Fuzzy Logic (Mamdani), PLC, Ladder Logic, control systems
Programming: C, C++, Python, Kotlin, JavaScript, TypeScript, PHP, SQL (basic)
Mobile & Web: Android (Jetpack Compose), React, NextJS, Typescript, Tailwind CSS, Express, Django, Fast API, Postgresql
Machine Learning & Computer Vision: TensorFlow, Keras, OpenCV, YOLO, CNN, transfer learning, Pandas, Matplotlib
Design & Tools: Fusion 360 (3D design and printing), Figma, Canva, Microsoft Office, Google Workspace
Soft Skills: Leadership, teaching and mentoring, teamwork, problem solving, time management